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GATE ECE · SIGNALS AND SYSTEMS · CHAPTER-WISE PYQS

Discrete Time Signal Z Transform

GATE ECE Signals and Systems · Previous Year Questions with Answer Key

37	26	1990 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	21 questions
Marks 2	16 questions
Answer key	at the end

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Marks 1 21 questions

Q1. GATE ECE 2026

The response of a discrete time system $\mathrm{y}[\mathrm{n}]$ obeys the following relation:

$y[n] = \frac{5}{6}y[n-1] - \frac{1}{6}y[n-2] + x[n]$.

The input to the system is $x[n] = \delta[n] - \frac{1}{3}\delta[n-1]$.

Which of the following options is TRUE for $y[n]$?

Open solution

(A) Stable and causal response

(B) Stable and non-causal response

(C) Unstable and causal response

(D) Unstable and non-causal response

Q2. GATE ECE 2025

Let $x[n]$ be a discrete-time signal whose z-transform is $X(z)$. Which of the following statements is/are TRUE?

Open solution

(A) The discrete-time Fourier transform (DTFT) of $x[n]$ always exists

(B) The region of convergence (RoC) of $X(z)$ contains neither poles nor zeros

(C) The discrete-time Fourier transform (DTFT) exists if the region of convergence (RoC) contains the unit circle

(D) If $x[n] = \alpha\delta[n]$, where $\delta[n]$ is the unit impulse and α is a scalar, then the region of convergence (RoC) is the entire z-plane

Q3. GATE ECE 2024

For a causal discrete-time LTI system with transfer function

$H(z) = \frac{2z^2+3}{(z+\frac{1}{3})(z-\frac{1}{3})}$

which of the following statements is/are true?

Open solution

(A) The system is stable.

(B) The system is a minimum phase system.

(C) The initial value of the impulse response is 2.

(D) The final value of the impulse response is 0.

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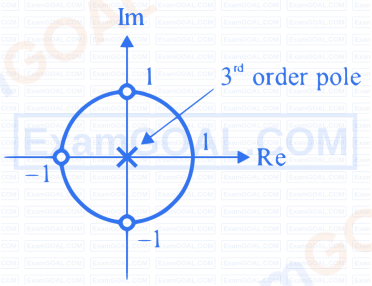
Q4. GATE ECE 2020

Open solution

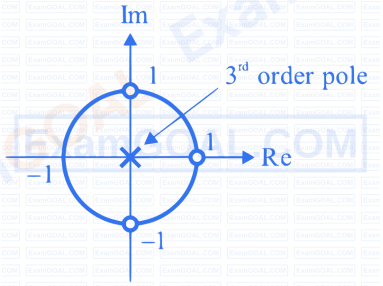
Which one of the following pole-zero plots corresponds to the transfer function of an LTI system characterized by the input-output difference equation given below?

$y[n] = \sum_{k=0}^{3}(-1)^k x[n-k]$

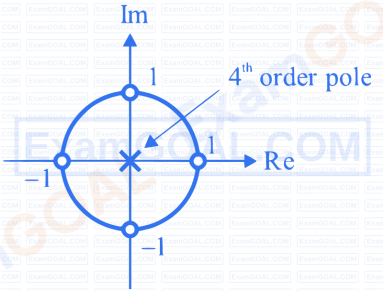
(A)



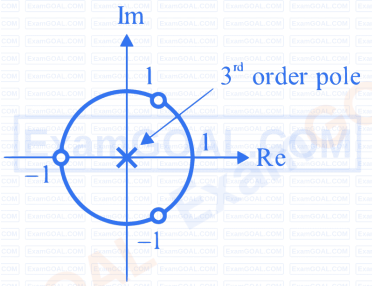
(B)



(C)



(D)



Q5. GATE ECE 2018

Open solution

A discrete time all-pass system has two of its poles at $0.25\angle 0^\circ$ and $\angle 30^\circ$. Which one of the following statements about the system is TRUE?

(A) It has two more poles at $0.5\angle 30^\circ$ and $4\angle 0^\circ$.

(B) It is stable only when the impulse response is two-sided.

(C) It has constant phase response over all frequencies.

(D) It has constant phase response over the entire z-plane.

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Q6. GATE ECE 2016 Set 1

Consider the sequence
$x[n] = a^n u[n] + b^n u[n]$, where u[n] denotes the unit step sequence and $0<|a|<|b|<1$.
The region of convergence (ROC) of the z-transform of $x[n]$ is

(A) $|z| > |a|$
(B) $|z| > |b|$
(C) $|z| < |a|$
(D) $|a| < |z| < |b|$

Open solution

Q7. GATE ECE 2016 Set 3

A discrete-time signal$x[n] = \delta[n-3] + 2\delta[n-5]$ has z-transform x(z). If Y (z)=X (-z) is the z-transform of another signal y$[n]$, then

(A) $y[n] = x[n]$
(B) $y[n] = x[-n]$
(C) $y[n] = -x[n]$
(D) $y[n] = -x[-n]$

Open solution

Q8. GATE ECE 2015 Set 2

Two casual discrete-time signals $x[n]$ and $y[n] = \sum_{m=0}^{n} x[m]$. If the z-transform of y$[n]$=$\frac{2}{z(z-1)^2}$, the value of $x[2]$ is _____

Open solution

Q9. GATE ECE 2014 Set 4

The sequence x $[n]$ = 0.5^n u[n], where u$[n]$ is the unit step sequence, is convolved with itself to obtain y $[n]$. Then $\sum_{n=\infty}^{+\infty} y[n]$ is _____.

Open solution

Q10. GATE ECE 2014 Set 2

An FIR system is described by the system function

$$H(z) = 1 + \frac{7}{2}z^{-1} + \frac{3}{2}z^{-2}$$

(A) maximum phase (B) manimum phase (C) mixed phase (D) zero phase

Open solution

Q11. GATE ECE 2014 Set 2

Let x$[n]$ = x$[-n]$. Let X(z) be the z-transform of x$[n]$. if 0.5 +j 0.25 is a zero of X(z), which one of the folowing must also be a zero of X (z)

(A) 0.5 - j0.25
(B) 1/ (0.5 + j0.25)
(C) 1/ (0.5 - j 0.25)
(D) 2 + j4

Open solution

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Q12. GATE ECE 2014 Set 3

For an all-pass system H(z)= $\frac{(z^{-1}-b)}{(1-az^{-1})}$ where $|H(e^{-j\omega})| = 1$, for all ω. If Re (a) $\neq 0$,$\mathrm{Im}(a) \neq 0$ then b equals

(A) a (B) a* (C) 1/a* (D) 1/a

Open solution

Q13. GATE ECE 2012

If $x\,[n]$= $(1/3)^{|n|} - (1/2)^n u\,[n]$, then the region of convergence (ROC) of its Z- transform in the Z-plane will be

(A) $\frac{1}{3} < |z| < 3$ (B) $\frac{1}{3} < |z| < \frac{1}{2}$

(C) $\frac{1}{2} < |z| < 3$ (D) $\frac{1}{3} < |z|$

Open solution

Q14. GATE ECE 2010

Consider the z-transform
X(z)=$5z^2 + 4z^{-1} + 3; 0 < |z| < \infty$.
The inverse z - transform x $[n]$ is

(A) $5\,\delta[n+2] + 3\,\delta\,[n] + 4\delta[n-1]$ (B) $5\,\delta[n-2] + 3\,\delta[n] + 4\,\delta[n+1]$

(C) $5\,u[n+2] + 3\,u[n] + 4\,u[n-1]$ (D) $5\,u[n-2] + 3\,u[n] + 4\,u[n+1]$

Open solution

Q15. GATE ECE 2009

The ROC of Z-transform of the discrete time sequence
x(n)= $\left(\frac{1}{3}\right)^n u(n) - \left(\frac{1}{2}\right)^n u(-n-1)$ is

(A) $\frac{1}{3} < |z| < \frac{1}{2}$ (B) $|z| > \frac{1}{2}$

(C) $|z| < \frac{1}{2}$ (D) $2 < |z| < 3$

Open solution

Q16. GATE ECE 2006

If the region of convergence of $x_1\,[n]$ + $x_2\,[n]$ is 1/3< $|z|$<2/3, then the region of convergence of $x_1\,[n]$ - $x_2\,[n]$ includes

(A) $\frac{1}{3} < |z| < 3$ (B) $\frac{2}{3} < |z| < 3$

(C) $\frac{3}{2} < |z| < 3$ (D) $\frac{1}{3} < |z| < \frac{2}{3}$

Open solution

Q17. GATE ECE 2005

The region of convergence of z-transform of the sequence $\left(\frac{5}{6}\right)^n u(n) - \left(\frac{6}{5}\right)^n u(-n-1)$ must be

(A) $|z| < \frac{5}{6}$ (B) $|z| > \frac{6}{5}$

(C) $\frac{5}{6} < |z| < \frac{6}{5}$ (D) $\frac{6}{5} < |z| < \infty$

Open solution

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Q18. GATE ECE 2004

The z transform of a system is
H(z) = $\frac{z}{z-0.2}$.
If the ROC is $|z|$ < 0.2, then the impulse response of the system is

(A) $(0.2)^n\, u\,[n]$ (B) $(0.2)^n\, u\,[-n-1]$
(C) $-(0.2)^n\, u\,[n]$ (D) $-(0.2)^n\, u\,[-n-1]$

Open solution

Q19. GATE ECE 2001

The region of convergence of the z- transform of a unit step function is

(A) $|z| > 1$ (B) $|z| < 1$
(C) (Real part of z)> 0 (D) (Real part of z)< 0

Open solution

Q20. GATE ECE 1999

The z-transform F(z) of the function f(nT) = a^{nT} is

(A) $\frac{z}{z-a^T}$ (B) $\frac{z}{z+a^T}$ (C) $\frac{z}{z-a^{-T}}$ (D) $\frac{z}{z+a^{-T}}$

Open solution

Q21. GATE ECE 1998

The z - transform of the time function $\sum_{k=0}^{\infty} \delta\,(n-k)$ is

(A) $(z-1)/z$ (B) $z/(z-1)^2$ (C) $z/(z-1)$ (D) $(z-1)^2/z$

Open solution

Marks 2 16 questions

Q22. GATE ECE 2020

The transfer function of a stable discrete - time LTI system is $H(z) = \frac{K(z-\alpha)}{(z+0.5)}$ where K and α are real numbers. The value of α (rounded off to one decimal place) with $|\alpha| > 1$, for which magnitude response of the system is constant over all frequencies, is ____ .

Open solution

Q23. GATE ECE 2016 Set 1

A sequence x$[n]$ is specified as $\begin{bmatrix} x\,[n] \\ x\,[n-1] \end{bmatrix} = \begin{bmatrix} 1\,1 \\ 1\,0 \end{bmatrix}^n \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, for n $\geq$2.
The initial conditions are x$[0]$ = 1, x$[1]$=1 and x$[n]$=0 for n< 0. The value of x$[12]$ is
_____.

Open solution

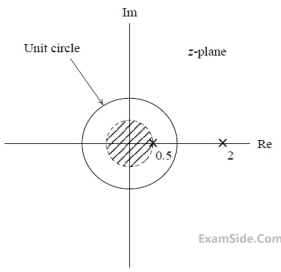
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Q24. GATE ECE 2016 Set 3

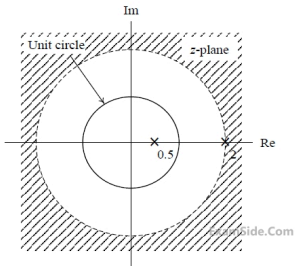
Open solution

The ROC (region of convergence) of the z-transform of a discrete-time signal is represented by the shaded region in the z-plane. If the signal $x[n] = (2.0)^{|n|}$, $-\infty < n < +\infty$ then the ROC of its z-transform is represented by

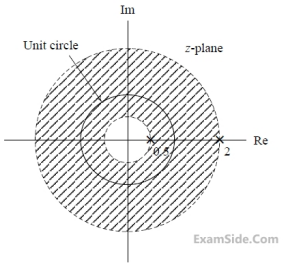
(A)



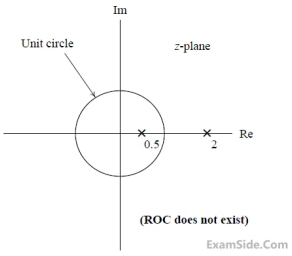
(B)



(C)



(D)

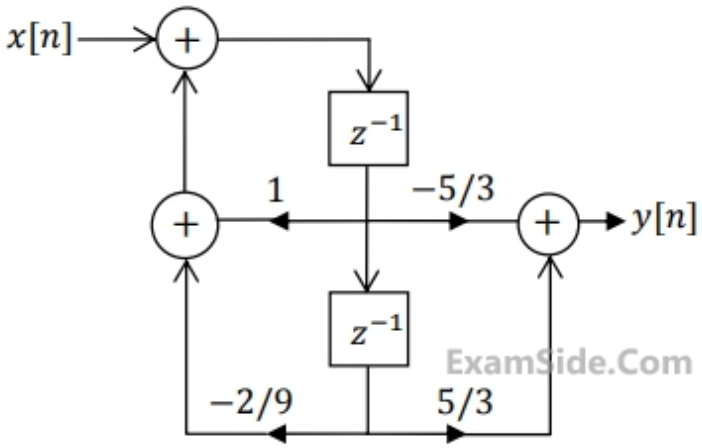


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Q25. GATE ECE 2015 Set 3

Open solution

A realization of a stable discrete time system is shown in the figure. If the system is excited by a unit step sequence input x[n] , the response y[n] is



(A) $4\left(-\frac{1}{3}\right)^n u[n] - 5\left(-\frac{2}{3}\right)^n u[n]$

(B) $5\left(-\frac{2}{3}\right)^n u[n] - 3\left(-\frac{1}{3}\right)^n u[n]$

(C) $5\left(\frac{1}{3}\right)^n u[n] - 5\left(\frac{2}{3}\right)^n u[n]$

(D) $5\left(\frac{2}{3}\right)^n u[n] - 5\left(\frac{1}{3}\right)^n u[n]$

Q26. GATE ECE 2015 Set 3

Open solution

Suppose x $[n]$ is an absolutely summable discrete-time signal. Its z-transform is a rational function with two poles and two zeroes. The poles are at z = ± 2j. Which one of the following statements is TRUE for the signal x=$[n]$?

(A) It is a finite duration signal.

(B) It is a causal signal.

(C) It is a non-causal signal.

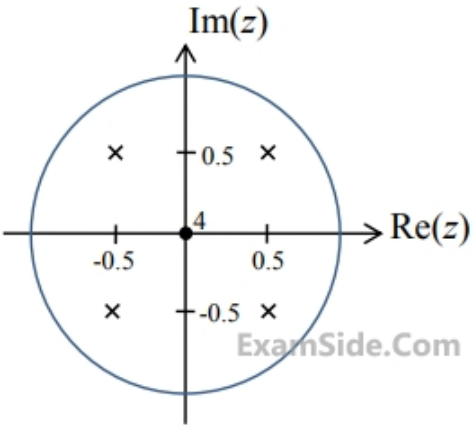
(D) It is a periodic signal.

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Q27. GATE ECE 2015 Set 1

The pole-zero diagram of a causal and stable discrete-time system is shown in the figure. The zero at the origin has multiplicity 4. The impulse response of the system is *h*[n]. If *h*[0] =1, we can conclude.

Open solution



(A) h (n) is real for all n.

(B) h (n) is purely imaginary for all n.

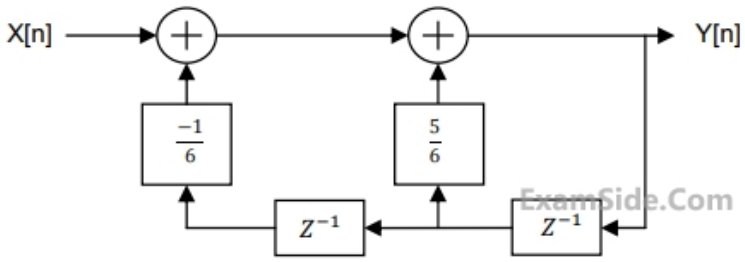
(C) h (n) is real for only even n.

(D) h (n) is purely imaginary for only odd n *

Q28. GATE ECE 2015 Set 1

For the discrete-time system shown in the figure, the poles of the system transfer function are located at

Open solution



(A) 2, 3 (B) 1/2, 3 (C) 1/2 , 1/3 (D) 2, 1/3

Q29. GATE ECE 2014 Set 1

Let x $[n]$= $\left(-\frac{1}{9}\right)^n u(n) - \left(-\frac{1}{3}\right)^n u(-n-1)$. The region of Convergence (ROC) of the z-tansform of x$[n]$

Open solution

(A) is $|z| > \frac{1}{9}$ (B) is $|z| < \frac{1}{3}$

(C) is $\frac{1}{3} > |z| > \frac{1}{9}$ (D) does not exist.

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Q30. GATE ECE 2014 Set 2

Open solution

The input-output relationship of a causal stable LTI system is given as $y[n] = \alpha\, y[n-1] + \beta\, x[n]$.

If the impulse response h[n] of this system satisfies the condition $\sum_{n=0}^{\infty} h[n] = 2$, the relationship between α and is α and β is

(A) $\alpha = 1 - \frac{\beta}{2}$ (B) $\alpha = 1 + \frac{\beta}{2}$

(C) $\alpha = 2\beta$ (D) $\alpha = -2\beta$

Q31. GATE ECE 2014 Set 3

Open solution

The z-transform of the sequence $x[n]$ is given by $X(z) = \frac{1}{(1-2z^{-1})^2}$, with the region of convergence $|z| > 2$. Then, $x[2]$ is _____.

Q32. GATE ECE 2014 Set 3

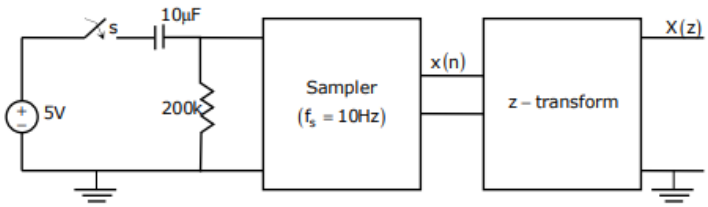
Open solution

Let $H_1(z) = (1 - pz^{-1})^{-1}$, $H_2(z) = (1 - qz^{-1})^{-1}$, $H(z) = H_1(z) + r\, H_2$. The quantities p, q, r are real numbers. Consider , $p=\frac{1}{2}$, $q=-\frac{1}{4}$ $|r| < 1$. If the zero H(z) lies on the unit circle, the r = _____.

Q33. GATE ECE 2008

Open solution

In the following network (Fig.1), the switch is closed at t = 0 and the sampling starts from t=0. The sampling frequency is 10 Hz.



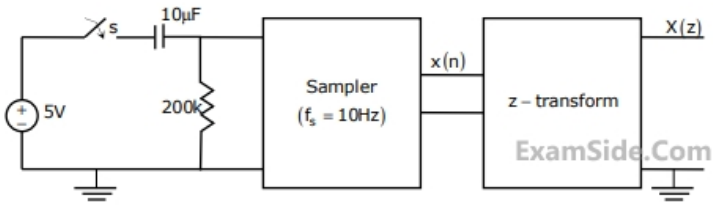
The samples x (n) (n=0, 1, 2,.....) are given by

(A) $5(1-e^{-0.05n})$ (B) $5e^{-0.05n}$ (C) $5(1-e^{-5n})$ (D) $5e^{-5n}$

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Q34. GATE ECE 2008

In the following network (Fig .1), the switch is closed at t = 0- and the sampling starts from t = 0. The sampling frequency is 10 Hz.



The expression and the region of convergence of the z-transform of the sampled signal are

(A) $\frac{5z}{z-e^{-5}}, |z| < e^{-5}$

(B) $\frac{5z}{z-e^{-0.05}}, |z| < e^{-0.05}$

(C) $\frac{5z}{z-e^{-0.05}}, |z| > e^{-0.05}$

(D) $\frac{5z}{z-e^{-5}}, |z| < e^{-5}$

Open solution

Q35. GATE ECE 2007

The z-transform X (z) f a sequence x$[n]$ is given by = $\frac{0.5}{1-2z^{-1}}$. It is given that the region of convergence of X$[z]$ includes the unit circle. The value of x$[0]$ is

(A) -0.5 (B) 0 (C) 0.25 (D) 0.5

Open solution

Q36. GATE ECE 1999

The z-transform of a signal is given by c(z)=$\frac{1}{4}\frac{z^{-1}(1-z^{-4})}{(1-z^{-1})^2}$. Its final value is

(A) 1/4 (B) zero (C) 1.0 (D) infinity

Open solution

Q37. GATE ECE 1990

The Z-transform of the following real exponential sequence:
x(nT) = a^n, nT $\geq$ 0
=0, nT<0, a> 0
gives us by

(A) $\frac{1}{1-z^{-1}}; |z| > 1$

(B) $\frac{1}{1-az^{-1}}; |z| > a$

(C) 1 for all z

(D) $\frac{1}{1-az^{-1}}; |z| < a$

Open solution

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Answer Key

Marks 1

1. **A**	2. **C, D**	3. **A, C, D**	4. **B**	5. **B**
6. **B**	7. **C**	8. **0**	9. **4**	10. **C**
11. **B**	12. **B**	13. **C**	14. **A**	15. **A**
16. **D**	17. **C**	18. **D**	19. **A**	20. **A**
21. **C**				

Marks 2

22. **-2**	23. **233**	24. **D**	25. **C**	26. **C**
27. **A**	28. **C**	29. **C**	30. **A**	31. **12**
32. **0.5**	33. **B**	34. **C**	35. **B**	36. **C**
37. **B**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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